CONFIDENTIAL

SMART MAIZE FARMING SYSTEM

IoT + AI FOR RWANDAN SMALLHOLDER FARMERS

PRESENTED BY

Smart Maize Team

DATE

June 2026



IMG_REF_01

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Executive Agenda

01 Problem & Market Context

Rwanda agricultural challenges and TAM analysis.

02 Solution Architecture

IoT sensors, AI analytics, multi-channel delivery, expert workflow.

03 Business Model & Impact

Tiered subscriptions, 3-year revenue projection, SDG alignment.

04 Pilot Case Study

Eastern Province deployment across 500 partner farms.

05 Roadmap & Investment

5-phase scaling plan and $250K seed funding ask.

BUILD STATUS

BACKEND 100% // 29 TEST SUITES PASSING

Problem & Market Context RWANDA_AGRI_2026

Agricultural Crisis

Rwanda agriculture employs 70% of the population and contributes 31% of GDP, yet maize farmers lose 30-50% of potential yield to climate variability, Fall Armyworm infestation, and lack of real-time data.

⚠ FAW Yield Loss: $30-100M/yr across East Africa

↗ Smart Farming TAM: +18% CAGR



RWANDA TAM

$45M

↑ 18% CAGR

Solution Architecture CORE_PILLARS

01

IoT Sensor Network

Low-cost ruggedized sensors monitor soil moisture, temperature, humidity, and NPK levels across partner farms. GSM modules stream real-time data to the cloud API.

COMPLETE

02

AI Analytics Engine

Google Gemini 2.5 Flash analyzes sensor data, weather, and crop growth stage to deliver irrigation, fertilization, and Fall Armyworm detection recommendations at 92% accuracy.

PRODUCTION

03

Multi-Channel Delivery

Recommendations reach every farmer via Web dashboard, Mobile app, USSD menus (*123#), SMS alerts, and Voice assistant in Kinyarwanda, English, and French.

IN PROGRESS

KEY TAKEAWAY: End-to-end pipeline validated with 29 backend test suites.

WATER -35% / YIELD +25-40%

Business Model & Revenue

Y1-Y3_PROJECTION

YEAR 3 REVENUE

$1.03M

▲ 22x vs Year 1

MONTHLY OPEX @ 1K USERS

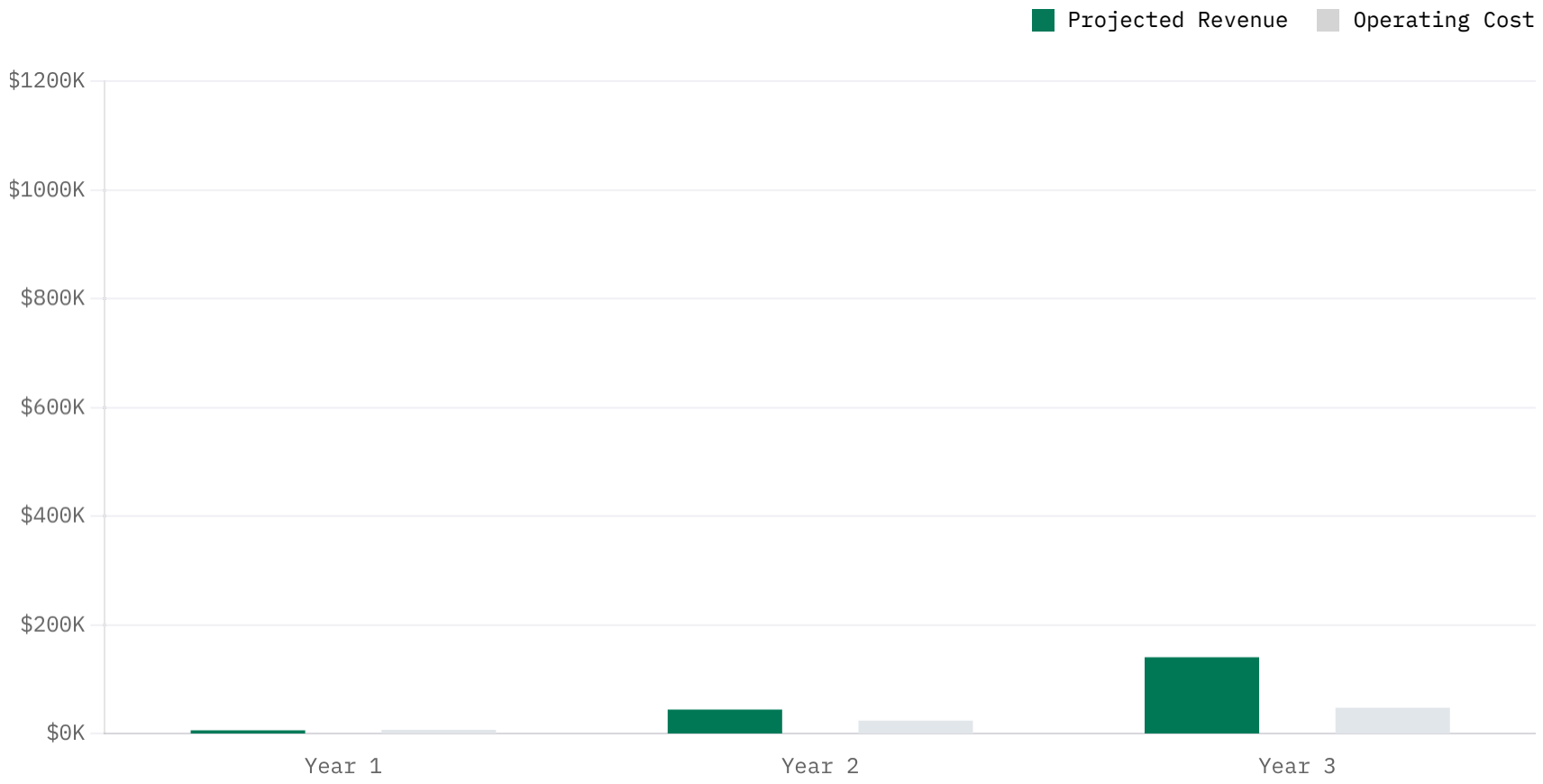
$4.5K

▼ Lean operations

YEAR 3 GROSS MARGIN

70%+

Target: 65%



User Roles & System Metrics

RBAC_&_PERFORMANCE

ROLE	ACCESS LEVEL	KEY ACTIONS	STATUS
Farmer	Owner	View sensors, receive recs, upload pest images, USSD/SMS access	OPTIMAL
Agric. Expert	Reviewer	Multi-farm monitor, validate AI recs, pest review, consulting	OPTIMAL
Admin	System	User mgmt, device tokens, audit logs, broadcast, analytics	STABLE
IoT Device	Ingest	HMAC-signed sensor data ingestion with anomaly detection	AUDITED

* RBAC enforced at Convex DB row level. JWT + Clerk authentication.



Test Suite Coverage

29 / 29

Backend integration tests passing

CONCURRENT WS

10K+

API THROUGHPUT

5K rps

Pilot Highlight: Eastern Province

CASE_STUDY_01



PILOT: BUGESERA, NYAGATARE, KIREHE

Timeline: Q2-Q3 2026 // 500 farms

THE CHALLENGE

Smallholder farmers face 30-50% yield losses from FAW infestation and over-irrigation, with no real-time soil data and limited expert access (1 officer per 3,000 farmers).

THE SOLUTION

Deployed IoT sensor kits across 500 farms, AI-powered irrigation scheduling via Gemini, USSD-delivered pest alerts in Kinyarwanda, and expert-in-the-loop FAW verification workflow.

WATER SAVED	YIELD GAIN	FAW ACCURACY
35%	+25-40%	92%

Roadmap & Risk Mitigation

01 Pilot Deployment

Deploy 500 IoT sensor kits across Bugesera, Nyagatare, and Kirehe districts. Onboard 1,000 farmers and 3 cooperatives. Validate AI recommendations in real field conditions.

! PRIORITY: HIGH // Q2-Q3 2026

02 Regional Expansion

Scale to 5 districts and 5,000 farmers in Year 1. Launch native Android app and cooperative plans. Enter Burundi border regions and begin East African Community rollout in Year 2.

TIMELINE: Q4 2026 - Q2 2028

03 Tech Innovation

Integrate satellite NDVI imagery for crop health at scale. Deploy ML models for 2-week FAW outbreak prediction. Build blockchain-based farm-to-market traceability and carbon credit tracking.

BUDGET APPROVED

04 Risk Mitigation

USSD-first strategy ensures access for the 54% of farmers without smartphones. Ruggedized IoT sensors with 12-month warranty. Privacy-by-design aligned with Rwanda Law N°058/2021.

CONTINUOUS MONITORING

EXECUTION READY

"Our platform is no longer a prototype - it is a working multi-service build with 29 passing test suites. With $250K in seed funding, we will deploy across 500 farms in Q2 2026 and prove the model at scale."

$250K	5,000	8-12x
SEED ASK	FARMERS // YEAR 1	5-YEAR ROI

/// SESSION END

Q&A
Contact

Full technical documentation, deployment runbook, and financial model available in the secure repository.

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ACCESS CODE
SM-AGRITECH-2026